Embedded AI midterm, TI-82 sheet

Midterm 2026-10-29, 18:00. Sources: lecture slides 1 to 6, the lecture 6 backpropagation example, practice series 1 to 3. Lines marked *general reminder* are not on the slides.

1. How to type it

- Each box below is one program: `PRGM`, `NEW`, type the name (8 characters max), `ENTER`, then one line per row. `2nd ALPHA` locks letters.
- Space: `ALPHA 0`. Equals sign: `2nd TEST 1`. Keep `LN`, `e^`, ^{2}, $\sqrt{}$ as keys, not letters.
- To read: `PRGM`, `EDIT`, pick the name. The editor shows long lines in full. Never run these programs (`EXEC`): they only hold text.
- Priority if time runs short: BACKPROP, METRIC, INFO, ACTIV, then the rest.

2. Programs to type

PRGM BACKPROP

```
E=1/2(D-Y)^2 COURSE 2026
S=SUM WX X0=1 BIAS
Y=1/(1+e^(-S))
DO=(D-Y)*Y(1-Y)
DH=DO*WOUT*H(1-H)
WNEW=W+A*DELTA*INPUT
INPUT OF BIAS=1
E=(D-Y)^2: DELTA*2
DH USES OLD WOUT
Y<D AND W>0: W UP
EX S1: W01=.8008
WITH *2: .8017
```

PRGM METRIC

```
POS=CLASS SEARCHED
TP=POS FOUND
FN=POS MISSED
FP=NEG FLAGGED POS
SENS=TP/(TP+FN)
SPEC=TN/(TN+FP)
ACC=(TP+TN)/TOT
PREC=TP/(TP+FP)
FPR=1-SPEC
ROC X=FPR Y=SENS
AUC .5 RANDOM 1 PERFECT
AUC<.5 WORSE THAN RANDOM
POS BELOW LIM:
FN=POS ABOVE
TP=ALLPOS-FN
TN=NEG ABOVE
FP=ALLNEG-TN
CHECK SUM=TOTAL
```

PRGM INFO

```
I=-PLOG2P-NLOG2N
LOG2X=LN X/LN 2
I(.5)=1 I(1)=0
I(.25)=.811
I(1/3)=.918
I(.4)=.971
I(.2)=.722
I(.3)=.881 I(.1)=.469
GAIN=I0-SUM(NK/N)IK
PURE BRANCH I=0
INSTANCES=PRODUCT
OF NB OF VALUES
N BIN ATTR FUNCTIONS
=2^(2^N)
```

PRGM ACTIV

```
SIG 0 TO 1 SIG(0)=.5
DSIG=Y(1-Y) MAX .25
TANH -1 TO 1
DTANH=1-Y^2 MAX 1
RELU=MAX(0,S)
DRELU=0 OR 1
SIGN PERCEPTRON +-1
VANISH: DSIG PRODUCT
SHRINKS OVER LAYERS
SIG AND TANH HIT
RELU AVOIDS IT
NO NONLINEAR ACT
=ONE LAYER POWER
```

PRGM DIST

```
EUCL=√(SUM DX^2) L2
MANH=SUM ABS DX L1
MINK P=1 MANH P=2 EUCL
COS=A.B/(NA*NB) TEXT
HAMMING=NB DIFF
BITS CATEGORICAL
MAHA USES VAR+CORR
2D:√((DX/SX)^2+(DY/SY)^2)
STD VERY DIFF: MAHA
```

PRGM CLUSTER

```
KMEANS: ASSIGN NEAREST
DIST^2 ENOUGH TO COMPARE
NEW C=MEAN OF GROUP
DEPENDS ON INIT
HIER CUT AT H:
NB CLUSTERS=NB OF
VERT LINES CROSSED
SINGLE=MIN COMPL=MAX
NO K NEEDED BEFORE
PCA MAX VARIANCE
LINEAR ONLY
```

PRGM GATES

```
LINE W0+W1X+W2Y=0
Y=-(W0+W1X)/W2
AND -1.5 1 1
OR -.5 1 1
NAND 1.5 -1 -1
XOR NOT LIN SEPARABLE
NEEDS HIDDEN LAYER
XOR=AND(OR,NAND)
```

```
PRGM EMBED
DRAM P=SUM CV²F/2
FASTER: MORE POWER
EDGE AI=ON DEVICE
BRAIN 1MS 20W
PARALLEL NOT SPEED
SIMPLE UI OR NONE
L1=SUM ABS W
L2=SUM W²
QUANT=LOW PRECISION
PRUNE=REMOVE WEIGHTS
FALL: ACCEL 3 AXES
62.5HZ TINY ML
IDLE WALK FALL
```

GATES weights and the XOR decomposition are a general reminder: the slides only say that a sign perceptron gives a linear separating surface. L1 and L2 regularization come from lab 2, not from the lecture slides.

3. Graphs: activation functions

Y= editor. Y1 is typed with 2nd Y−VARS, 1, 1. To show only some curves, put the cursor on the = of a line and press ENTER to switch it off.

Line	Type	Curve
Y1	1/(1+e^(−X))	sigmoid
Y2	Y1(1−Y1)	sigmoid derivative, bell, peak 0.25 at 0
Y3	(e^(X)−e^(−X))/(e^(X)+e^(−X))	tanh
Y4	1−Y3²	tanh derivative, peak 1 at 0
Y5	max(0,X) (MATH, NUM)	ReLU
Y6	(X>0) (2nd TEST)	ReLU derivative, 0 then 1

WINDOW: Xmin=−4 Xmax=4 Xscl=1 Ymin=−1.5 Ymax=1.5 Yscl=.5, then GRAPH. For the values to copy onto the exam grid: 2nd TblSet, TblMin=−3, ΔTbl=1, then 2nd TABLE. To evaluate at one point: .6 STO→ X, ENTER, then Y1 ENTER.

s	−3	−2	−1	0	1	2	3
sigmoid	0.047	0.119	0.269	0.5	0.731	0.881	0.953
sigmoid’	0.045	0.105	0.197	0.25	0.197	0.105	0.045
tanh	−0.995	−0.964	−0.762	0	0.762	0.964	0.995
tanh’	0.010	0.071	0.420	1	0.420	0.071	0.010

Entropy as a function. Y7=−(X LN(X)+(1−X)LN(1−X))/LN 2. Store the share of positives in X (.25 STO→ X), then Y7 ENTER gives I (0.811). Avoid $X = 0$ or 1: the answer is 0, the calculator gives an error.

4. Graphs: logic gates with a perceptron

The perceptron draws the line $w_0 + w_1 x_1 + w_2 x_2 = 0$. Plot the four inputs and the line, then read which corners fall on each side.

1. STAT, EDIT: L1={0,0,1,1}, L2={0,1,0,1}.
2. 2nd STAT PLOT, Plot1: On, scatter, Xlist L1, Ylist L2.
3. Store the weights: −1.5 STO→ A, 1 STO→ B, 1 STO→ C (here AND).
4. Y8=−(A+BX)/C, all other Y switched off.
5. WINDOW: Xmin=−.5 Xmax=1.5 Ymin=−.5 Ymax=1.5, then GRAPH.

Reading: AND leaves only $(1,1)$ above the line, OR leaves only $(0,0)$ below. For XOR, no choice of A, B, C puts $(0,1)$ and $(1,0)$ on one side and $(0,0)$, $(1,1)$ on the other: one line cannot separate it, a hidden layer is needed. Weights are a general reminder, see the note under the programs.